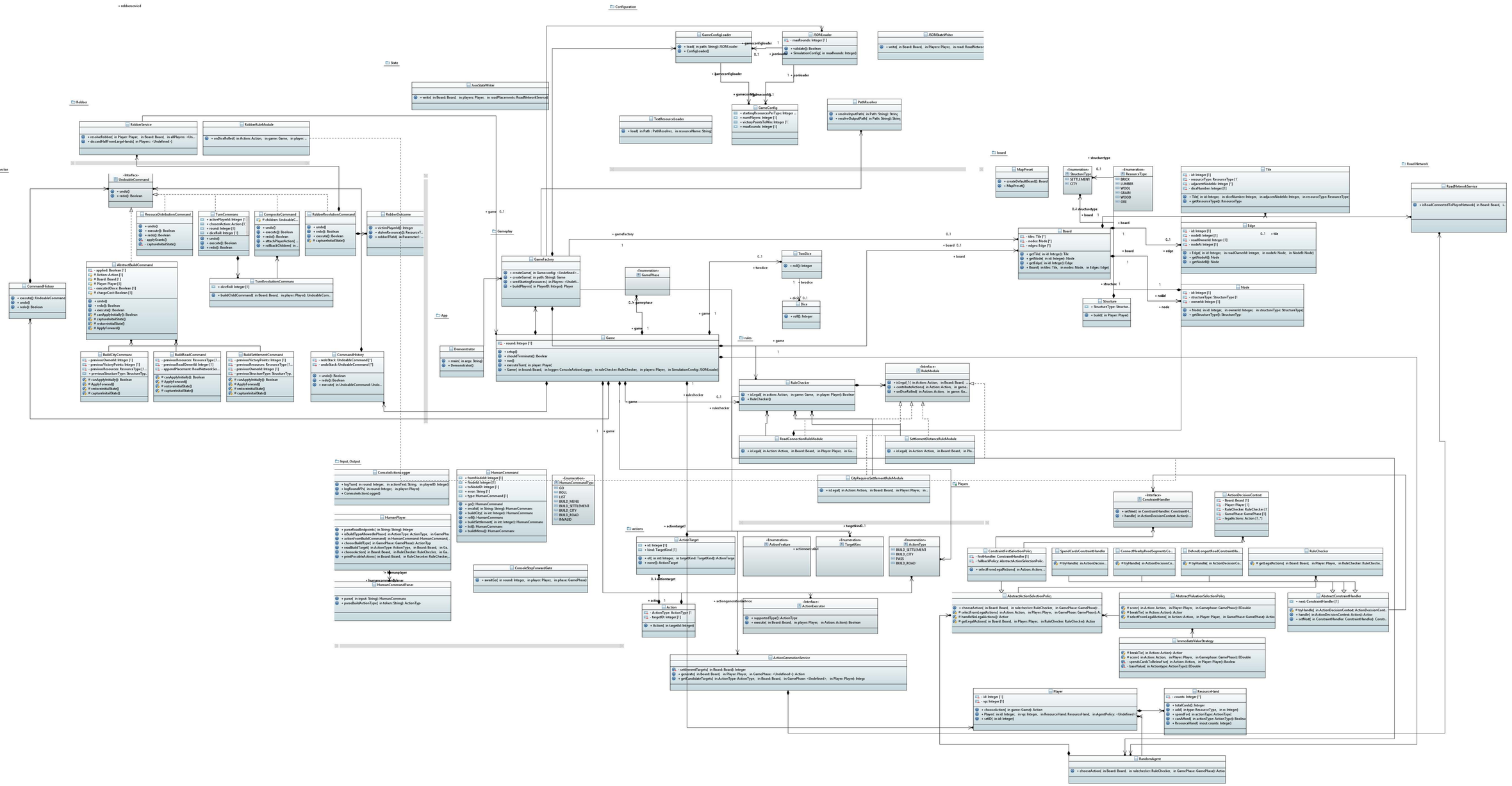
robberservice
Configuration
GameConfigLoader
+ load(in path: String): JSONLoader
+ ConfigLoader()
JSONLoader
- maxRounds: Integer [1]
+ validate(): Boolean
+ SimulationConfig(in maxRounds: Integer)
JSONStateWriter
+ write(in Board: Board, in Players: Player, in road: RoadNetwo
+ gameconfigloader
+ jsonloader
State
JsonStateWriter
+ write(in Board: Board, in players: Player, in roadPlacements: RoadNetworkService)
+ gameconfigloader
+ jsonloader
+ gameconfigloader
+ jsonloader
GameConfig
+ startingResourcesPerType: Integer ...
+ numPlayers: Integer [1]
+ victoryPointsToWin: Integer [1]
+ maxRounds: Integer [1]
TextResourceLoader
+ load(in Path : PathResolver, in resourceName: String)
PathResolver
+ resolveInputPath(in Path: String): String
+ resolveOutputPath(in Path: String): String
Robber
RobberService
+ resolveRobber(in Player: Player, in Board: Board, in allPlayers: <Un...
+ discardHalfFromLargeHands(in Players: <Undefined>)
RobberRuleModule
+ onDiceRolled(in Action: Action, in game: Game, in player: ...
Connector
-Interface-
UndoableCommand
+ undo()
+ redo(): Boolean
ResourceDistributionCommand
+ undo()
+ execute(): Boolean
+ redo(): Boolean
- applyGrants()
- captureInitialState()
TurnCommanc
+ activePlayerId: Integer [1]
+ chosenAction: Action [1]
+ round: Integer [1]
+ diceRoll: Integer [1]
+ undo()
+ execute(): Boolean
+ redo(): Boolean
CompositeCommand
children: UndoableC...
+ undo()
+ execute(): Boolean
+ redo(): Boolean
+ attachPlayerAction(...
+ rollBackChildren(in ...
RobberResolutionCommand
+ undo()
+ redo(): Boolean
+ execute(): Boolean
# captureInitialState()
RobberOutcome
+ victimPlayerId: Integer
+ stolenResource(): ResourceT...
+ robberTileId(in Parameter1): ...
AbstractBuildCommand
- applied: Boolean [1]
Action: Action [1]
Board: Board [1]
Player: Player [1]
- executedOnce: Boolean [1]
changeCost: Boolean [1]
+ undo()
+ redo(): Boolean
- execute(): Boolean
canApplyInitially(): Boolean
captureInitialState()
restoreInitialState()
ApplyForward()
TurnResolutionCommanc
+ diceRoll: Integer [1]
+ buildChildCommand(in Board: Board, in player: Player): UndoableCom...
CommandHistory
+ execute(): UndoableCommand
+ undo()
+ redo(): Boolean
BuildCityCommanc
- previousOwnerId: Integer [1]
- previousVictoryPoints: Integer [1]
- previousResources: ResourceType [1...
- previousStructureType: StructureTyp...
canApplyInitially(): Boolean
ApplyForward()
restoreInitialState()
captureInitialState()
BuildRoadCommand
- previousResources: ResourceType [1...
- previousRoadOwnerId: Integer [1]
- appendPlacement: RoadNetworkSer...
canApplyInitially(): Boolean
ApplyForward()
restoreInitialState()
captureInitialState()
BuildSettlementCommand
- previousVictoryPoints: Integer [1]
- previousResources: ResourceType [1...
- previousOwnerId: Integer [1]
- previousStructureType: StructureTyp...
canApplyInitially(): Boolean
ApplyForward()
restoreInitialState()
captureInitialState()
CommandHistory
- redoStack: UndoableCommand [*]
- undoStack: UndoableCommand [*]
+ undo(): Boolean
+ redo(): Boolean
+ execute(in UndoableCommand): Undo...
Gameplay
GameFactory
+ createGame(in Gameconfig: <Undefined>...
+ createGame(in path: String): Game
+ seedStartingResources(in Players: <Undefi...
+ buildPlayers(in Players[]: Integer): Player
+ gamefactory
+ game
TwoDice
+ roll(): Integer
+ twodice
-Enumeration-
GamePhase
+ gamephase
Dice
+ roll(): Integer
+ dice
+ twodice
App
Demonstrator
+ main(in args: String)
+ Demonstrator()
Game
- round: Integer [1]
+ setup()
+ shouldTerminate(): Boolear
+ run()
+ executeTurn(in player: Player(
+ Game(in board: Board, in logger: ConsoleActionLogger, in rulechecker: RuleChecker, in players: Player, in SimulationConfig: JSONLoader
+ game
+ game
+ game
+ game
rules
RuleChecker
+ isLegal(in action: Action, in game: Game, in player: Player): Boolear
+ RuleChecker()
-Interface-
RuleModule
+ isLegal_1(in Action: Action, in Board: Board, ...
+ combineActions(in Action: Action, in game: ...
+ onDiceRolled(in Action: Action, in game: Ga...
+ rulechecker
+ rulechecker
ROadConnectionRuleModule
+ isLegal(in Action: Action, in Board: Board, in Player: Player, in Ga...
SettlementDistanceRuleModule
+ isLegal(in Action: Action, in Board: Board, in Pla...
CityRequiresSettlementRuleModule
+ isLegal(in Action: Action, in Board: Board, in Player: Player, in ...
Input_Output
ConsoleActionLogger
+ logTurn(in round: Integer, in actionText: String, in playerID: Integer)
+ logRoundVPs(in round: Integer, in player: Player)
+ ConsoleActionLogger()
HumanCommand
+ fromNodeId: Integer [1]
+ toNodeId: Integer [1]
+ toRoadID: Integer [1]
+ error: String [1]
+ type: HumanCommandTyp
+ get(): HumanCommand
+ setId(in String: String): HumanCommanc
+ buildCity(in int: Integer): HumanCommanc
+ roll(): HumanCommanc
+ buildSettlement(in int: Integer): HumanCommanc
+ list(): HumanCommanc
+ buildMenu(): HumanCommanc
-Enumeration-
HumanCommandTyp
GO
ROLL
LIST
BUILD_MENU
BUILD_SETTLEMENT
BUILD_CITY
BUILD_ROAD
INVALID
HumanPlayer
+ parseRoadEndpoints(in String: String): Integer
+ isBuildTypeAllowedInPhase(in ActionType: ActionType, in GamePha
+ actionFromBuildCommand(in HumanCommand: HumanCommand
+ chooseBuildType(in GamePhase: GamePhase): ActionTyp
+ readBuildTarget(in ActionType: ActionType, in Board: Board, in Ga...
+ chooseAction(in Board: Board, in RuleChecker: RuleChecker, in Ga...
+ printPossibleActions(in Board: Board, in RuleChecker: RuleChecker,...
+ humanparser
+ humancommandparser
HumanCommandParser
+ parse(in input: String): HumanCommanc
+ parseBuildActionType(in token: String): ActionTyp
ConsoleStepForwardGate
+ awaitGo(in round: Integer, in player: Player, in phase: GamePhase)
actions
ActionTarget
+ id: Integer [1]
+ kind: TargetKind [1]
+ ref(in int: Integer, in targetKind: TargetKind): ActionTarge
+ owner(): ActionTarget
+ actiontarget
+ actiontarget
-Enumeration-
ActionFeature
+ actionnecessitate
-Enumeration-
TargetKinc
-Enumeration-
ActionType
BUILD_SETTLEMENT
BUILD_CITY
PASS
BUILD_ROAD
Action
- ActionType: ActionType [1]
- targetID: Integer [1]
+ Action(in targetID: Integer)
+ action
+ actionexecutor
-Interface-
ActionExecutor
+ supportedType(): ActionType
+ execute(in Board: Board, in player: Player, in Action: Action): Boolean
+ actiongenerationservice
ActionGenerationService
- settlementTargets(in Board: Board): Integer
+ generate(in Board: Board, in Player: Player, in GamePhase: <Undefined>): Action
+ getCandidateTargets(in ActionType: ActionType, in Board: Board, in GamePhase: <Undefined>, in Player: Player): Integ
board
MapPreset
+ createDefaultBoard(): Board
+ MapPreset()
-Enumeration-
StructureType
SETTLEMENT
CITY
+ structuretype
-Enumeration-
ResourceType
BRICK
LUMBER
WOOL
GRAIN
WOOD
ORE
+ structuretype
+ board
Board
- tiles: Tile [*]
- nodes: Node [*]
- edges: Edge [*]
+ getTile(in id: Integer): Tile
+ getNode(in id: Integer): Node
+ getEdge(in id: Integer): Edge
+ Board(in tiles: Tile, in nodes: Node, in Edges: Edge)
+ board
+ board
+ board
+ board
Structure
+ StructureType: Structur...
+ build(in Player: Player)
+ structure
+ structure
Tile
- id: Integer [1]
- resourceType: ResourceType [1
- adjacentNodeIds: Integer [*]
- diceNumber: Integer [1]
+ Tile(in id: Integer, in diceNumber: Integer, in adjacentNodeIds: Integer, in resourceType: ResourceType
+ getResourceType(): ResourceTyp
Edge
- id: Integer [1]
- nodeA: Integer [1]
- roadOwnerId: Integer [1]
- nodeA: Integer [1]
+ Edge(in id: Integer, in roadOwnerId: Integer, in nodeA: Node, in NodeB: Node)
+ getNodeA(): Node
+ getNodeB): Node
+ tile
+ edge
Node
- id: Integer [1]
- structureType: StructureType [1]
- ownerId: Integer [1]
+ Node(in id: Integer, in ownerId: Integer, in structureType: StructureType)
+ getStructureType(): StructureTyp
+ nodeid
+ node
Road Network
RoadNetworkService
+ isRoadConnectedToPlayerNetwork(in Board: Board, i...
Players
-Interface-
ConstraintHandler
+ setNext(in ConstraintHandler: ConstraintH...
+ handle(in ActionDecisionContext: Action): ...
ActionDecisionContext
- Board: Board [1]
- Player: Player [1]
- RuleChecker: RuleChecker [1]
- GamePhase: GamePhase [1]
- legalActions: Action [1..*]
ConstraintFirstSelectionPolicy
- firstHandler: ConstraintHandler [1]
- fallbackPolicy: AbstractActionSelectionPolic...
+ selectFromLegalActions(in Action: Action, ...
SpendCardsConstraintHandler
tryHandle(in ActionDecision...
ConnectNearbyRoadSegmentsCo...
tryHandle(in ActionDecisionCo...
DefendLongestRoadConstraintHa...
tryHandle(in ActionDecisionCo...
RuleChecker
getLegalActions(in Board: Board, in Player: Player, in RuleChecker: RuleChecke...
AbstractActionSelectionPolicy
+ chooseAction(in Board: Board, in rulechecker: RuleChecker, in GamePhase: GamePhase): ...
# selectFromLegalActions(in Action: Action, in Player: Player, in GamePhase: GamePhase): A...
# handleNoLegalAction(): Action
# getLegalActions(in Board: Board, in Player: Player, in RuleChecker: RuleChecker): Action
AbstractValuationSelectionPolicy
score(in Action: Action, in Player: Player, in Gamephase: GamePhase): EDouble
breakTie(in Action: Action): Action
selectFromLegalActions(in Action: Action, in Player: Player, in GamePhase: GamePhase): Actio...
AbstractConstraintHandler
- next: ConstraintHandler [1]
tryHandle(in ActionDecisionContext: ActionDecisionCont...
+ handle(in ActionDecisionContext: Action): Actior
+ setNext(in ConstraintHandler: ConstraintHandler): Constr...
ImmediateValueStrategy
breakTie(in Action: Action): Action
score(in Action: Action, in Player: Player, in Gamephase: GamePhase): EDouble
- - spendCardsToBeforeFirst(in Action: Action, in Player: Player): Boolea
- - baseValue(in ActionType: ActionType): EDouble
Player
- id: Integer [1]
- vp: Integer [1]
+ chooseAction(in game: Game): Action
+ Player(in id: Integer, in vp: Integer, in resourceHand: ResourceHand, in AgentPolicy: <Undefined>)
+ setID(in id: Integer)
ResourceHand
- counts: Integer [*]
+ totalCards(): Integer
+ add(in type: ResourceType, in n: Integer)
+ spendFrom(in actionType: ActionType)
+ canAfford(in actionType: ActionType): Boolea
+ ResourceHand(inout counts: Integer)
RandomAgent
+ chooseAction(in Board: Board, in rulechecker: RuleChecker, in GamePhase: GamePhase): Actio